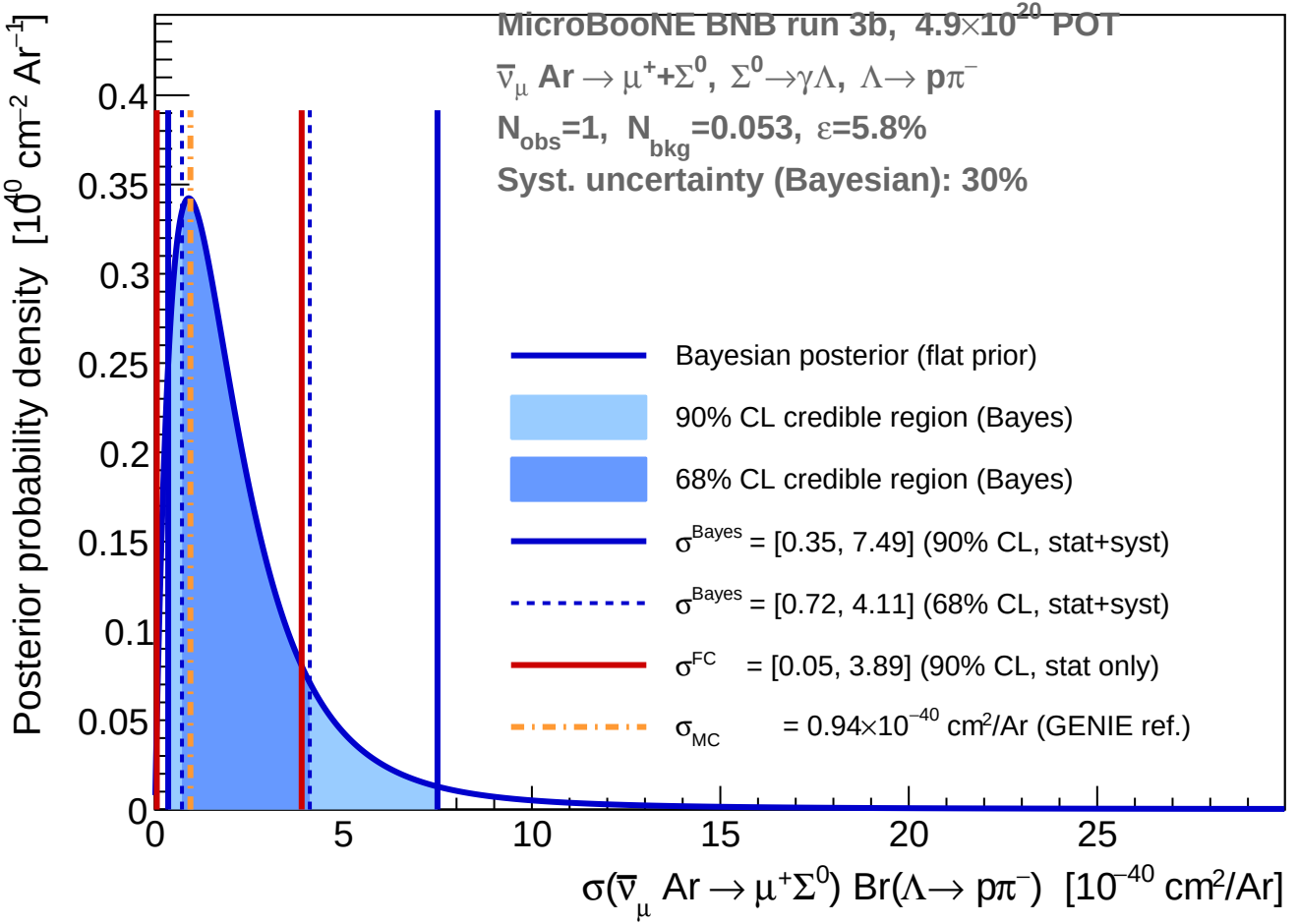
MicroBooNE BNB run 3b, 4.9×10^20 POT
ν̄μ Ar → μ+Σ0, Σ0→γΛ, Λ→ pπ−
Nobs=1, Nbkg=0.053, ε=5.8%
Syst. uncertainty (Bayesian): 30%
Bayesian posterior (flat prior)
90% CL credible region (Bayes)
68% CL credible region (Bayes)
σBayes = [0.35, 7.49] (90% CL, stat+syst)
σBayes = [0.72, 4.11] (68% CL, stat+syst)
σFC = [0.05, 3.89] (90% CL, stat only)
σMC = 0.94×10−40 cm2/Ar (GENIE ref.)
Posterior probability density [10^40 cm^−2 Ar^−1]
σ(ν̄μ Ar → μ+Σ0) Br(Λ→ pπ−) [10^−40 cm2/Ar]
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0
5
10
15
20
25